

Chapter

Mole Concept And Stoichiometry

1. The ratio between the volumes occupied by 4.4 grams of carbon dioxide and 2 grams of hydrogen gas is: SQP 2025

- (a) 22 : 1
- (b) 1 : 2.2
- (c) 1 : 10
- (d) 10 : 1

2. The number of atoms present in one molecule of an element is called its :

MAIN 2021

- (a) Molecular number
- (b) Atomic number
- (c) Atomicity
- (d) Avogadro's number

3. The empirical formula for benzene is : COMP 2023

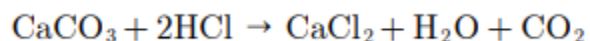
- (a) CH
- (b) C₂ H₂
- (c) C₄ H₈
- (d) C₆ H₆

4. Define atomicity of a gas. SQP 2023

5. How will you calculate percentage of an element in a compound ?

SQP 2023

6. Calcium carbonate reacts with dilute hydrochloric acid as given below:



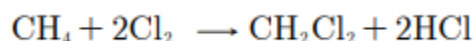
MAIN 2024

(a) What is the mass of 5 moles of calcium carbonate? (Relative molecular mass of calcium carbonate is 100)

(b) How many moles of HCl will react with 5 moles of calcium carbonate?

(c) What is the volume of carbon dioxide liberated at S.T.P. at the same time?

7. Calculate the volume of HCl gas formed and chlorine gas required when 40 mL of methane reacts completely with chlorine at S.T.P.



8. Calculate the percentage of water of crystallisation in hydrated copper sulphate $6 [\text{CuSO}_4 \cdot 5\text{H}_2\text{O}]$. COMP 2023

9. Chlorine has atomic weight 35.5. Give reasons. COMP 2021

10. Under the same conditions of temperature and pressure you collect 2 litres of carbon dioxide, 3 litres of chlorine, 5 litres of hydrogen, 4 litres of nitrogen and 1 litre of sulphur dioxide. In which gas sample will there be ?
MAIN 2021

(a) the greatest number of molecules.

(b) the least number of molecules.

11. A 2.5 g sample of hydrate of $\text{Ca}(\text{NO}_3)_2$ was heated and only 1.7 g of anhydrous salt remained. What percentage of water was in the hydrate ?
MAIN 2021

12. Washing soda has the formula $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$. What mass of anhydrous sodium carbonate is left when all the water of crystallization is expelled by heating 57.2 g of washing soda ?

COMP 2023

Solution

1. Thus (c) is correct option.

2. (c) Atomicity.

The number of atoms present in one molecule of an element is called its **atomicity**.

3. (d) is correct option.

The empirical formula is the simplest wholenumber ratio defining constituent atoms in a species. Thus for benzene, whose molecular formula is C_6H_6

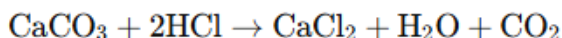
4. **Atomicity of a gas** is defined as the **number of atoms present in one molecule of the gas**.

5. To calculate the **percentage of an element in a compound**, use the formula:

$$\text{Percentage of an element} = \frac{\text{Total mass of the element in the compound}}{\text{Molecular mass of the compound}} \times 100$$

6.

Given reaction:



(a) Mass of 5 moles of $CaCO_3$

$$\text{Molar mass of } CaCO_3 = 100 \text{ g/mol}$$

$$\text{Mass} = \text{Number of moles} \times \text{Molar mass} = 5 \times 100 = 500 \text{ g}$$

✓ Mass of 5 moles of $CaCO_3 = 500 \text{ g}$

(b) Moles of HCl required

From the balanced equation, 1 mole of $CaCO_3$ reacts with 2 moles of HCl.

$$\text{Moles of HCl} = 5 \text{ moles of } CaCO_3 \times 2 = 10 \text{ moles}$$

✓ Moles of HCl = 10 moles

(c) Volume of CO_2 liberated at STP

1 mole of CaCO_3 produces 1 mole of CO_2 . So, 5 moles of CaCO_3 produce 5 moles of CO_2 .

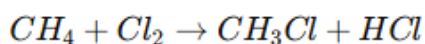
At STP, 1 mole of gas = 22.4 L.

$$\text{Volume of } \text{CO}_2 = 5 \times 22.4 = 112 \text{ L}$$

✓ Volume of $\text{CO}_2 = 112 \text{ L}$

7.

Step 1: Write the balanced equation



From the equation:

- 1 mole of CH_4 reacts with 1 mole of Cl_2 to produce 1 mole of HCl .
 - Volume relationship at STP: 1 volume of gas reacts with 1 volume of gas to give 1 volume of gas.
-

Step 2: Determine volumes

(a) Volume of chlorine gas required

- 1 volume of CH_4 reacts with 1 volume of Cl_2

$$V_{\text{Cl}_2} = V_{\text{CH}_4} = 40 \text{ mL}$$

(b) Volume of hydrogen chloride gas formed

- 1 volume of CH_4 gives 1 volume of HCl

$$V_{\text{HCl}} = V_{\text{CH}_4} = 40 \text{ mL}$$

✓ Step 3: Answer

- Volume of HCl formed = 40 mL
- Volume of Cl_2 required = 40 mL

8.

Molecular mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

$$= 63.5 + 32 + (4 \times 16) + 5(2 \times 1) + (16)$$

$$= 63.5 + 32 + 64 + 90 = 249.5 \text{ amu}$$

Percentage of water of crystallisation

$$= \frac{90}{249.5} \times 100 = 36.07\%$$

9. Chlorine has an **atomic weight of 35.5** because it exists naturally as a mixture of two isotopes:

- **Chlorine-35** (mass = 35 u, abundance $\approx 75\%$)
- **Chlorine-37** (mass = 37 u, abundance $\approx 25\%$)

The **average atomic mass** is calculated using their relative abundances:

$$\text{Atomic weight} = (35 \times 0.75) + (37 \times 0.25) = 26.25 + 9.25 = 35.$$

10. (a) 5 litres of hydrogen

(b) 1 litre of sulphur dioxide

(In accordance with the Avogadro's law).

11. **Given:**

- Mass of hydrate = 2.5 g
- Mass of anhydrous salt = 1.7 g

Step 1: Calculate mass of water lost

$$\begin{aligned} \text{Mass of water} &= \text{Mass of hydrate} - \text{Mass of anhydrous salt} \\ &= 2.5 - 1.7 = 0.8 \text{ g} \end{aligned}$$

Step 2: Calculate percentage of water in hydrate

$$\begin{aligned} \% \text{Water} &= \frac{\text{Mass of water}}{\text{Mass of hydrate}} \times 100 \\ &= \frac{0.8}{2.5} \times 100 = 32\% \end{aligned}$$

Answer:

32% water

12.

Given:

- Mass of washing soda = 57.2 g
- Formula: $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

Step 1: Calculate molar masses

- $\text{Na}_2\text{CO}_3 = 2(23) + 12 + 3(16) = 106 \text{ g/mol}$
- $10\text{H}_2\text{O} = 10 \times (2 + 16) = 180 \text{ g/mol}$
- Molar mass of $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O} = 106 + 180 = 286 \text{ g/mol}$

Step 2: Calculate mass fraction of anhydrous Na_2CO_3

$$\text{Mass fraction} = \frac{106}{286} \approx 0.3706$$

Step 3: Calculate mass of anhydrous Na_2CO_3 in 57.2 g of washing soda

$$\text{Mass of } \text{Na}_2\text{CO}_3 = 57.2 \times 0.3706 \approx 21.2 \text{ g}$$

Answer:

21.2 g
